

Not Incidental:

Congenital Anomalies as Wrapper-Failure Markers in Hereditary Cancer Predisposition

A positional paper

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Companion to McCarthy 2026 Paper 5 (Cancer eLife) and Paper 6 (ChannelStructure), and to Paper 8 (The Cost of Computation) and Paper 8b (Presence Without Performance). Stakes a clinically actionable position derived from the channel hierarchy framework, identifies the published evidence that already partially confirms it, names the specific test that has not been run, and lays out the clinical action that follows today.

Abstract

A bicornuate uterus discovered during obstetric workup. A horseshoe kidney detected incidentally on MRI. A repaired esophageal atresia. A small ventricular septal defect closed in childhood. Current practice treats these congenital anomalies as developmental history irrelevant to adult oncology. We argue the relationship is the opposite, with one specific testable claim leading: **women with Müllerian developmental anomalies (bicornuate uterus, septate uterus, didelphys, unicornate) show elevated rates of ER/PR-positive breast cancer at the population level, in distant tissue, beyond what can be explained by local-tissue cancer elevation, because the same molecular wrapper failure that produced the embryonic Müllerian anomaly later produces the adult breast cancer through the same Knudson second-hit mechanism in a different tissue at a different timescale.**

The prediction is a derivation from the channel hierarchy framework (Paper 5, Paper 6), not an extrapolation from any single patient. HR-repair / FA-pathway / wrapper genes form a specific channel class whose failure modes have been characterized in cross-species and within-tumor analyses. The novel claim of this paper is that those failure modes also fire during embryonic morphogenesis when the same molecular event (Knudson second-hit loss-of-heterozygosity at the second functional allele) occurs in cells doing morphogenetic work during a critical developmental window. The developmental anomaly is therefore not a coincidence — it is the same molecular event that the adult cancer is, just at an earlier life stage.

The basic population-level inverted test (“does any birth defect predict any adult cancer?”) has already been run by Daltveit et al 2020 BMJ, using Norwegian and Swedish national health registries, with overall OR 1.74 (99% CI 1.63-1.84) and adult-period OR 1.21 persisting after the strongest childhood signal. **This paper accepts the Daltveit result as the existing scaffold, and proposes three specific extensions** that have not been tested: (1) HBOC-spectrum tissue specificity in distant cancers, after controlling for the same-organ-system effect Daltveit reports; (2) ER/PR receptor subtype concentration of the breast cancer signal, predicting hormone-responsive HR-deficient cancers specifically; (3) cross-tissue lineage specificity (Müllerian anomaly → ER/PR+ breast cancer rather than just any breast cancer). All three are testable on the same Norwegian/Swedish registry infrastructure Daltveit used, with the addition of cancer-subtype data already present in those national registries.

The clinical action the framework supports does not require waiting for the test. The combined existing evidence — Daltveit OR 1.74, the bi-allelic Fanconi anemia → VACTERL-H literature, the horseshoe-kidney → 3-4× elevated cancer literature, the OpenTargets gene-disease aggregation showing HBOC genes are linked to specific developmental phenotypes via clinical-genetics submissions — is already enough to support precautionary recommendations: women with Müllerian developmental anomalies should be considered for earlier breast cancer screening; patients with multiple VACTERL-spectrum anomalies should be considered for HR-deficient gene panel testing; HBOC carriers with documented developmental anomalies should be moved to the high-intensity end of the surveillance interval recommended for their gene.

This paper does not present a new experiment. It is positional: it derives a sharper prediction from an existing framework, acknowledges what is already published, identifies what is not, names the collaboration target, and stakes the clinical action that follows.

1. The position

We claim three connected things:

Specific claim: Women with Müllerian developmental anomalies (ICD-10 Q51, Q52 — bicornuate uterus, septate uterus, didelphys, unicornate, vaginal anomalies) show elevated rates of ER/PR-positive breast cancer at the population level, beyond same-organ-system local-tissue cancer elevation, with effect size large enough to be clinically meaningful.

Generalized claim: Patients with congenital anomalies in the VACTERL or Müllerian spectrum show elevated rates of HR-deficient adult cancers in distant tissues, with elevation concentrated in HBOC-spectrum cancer types (breast, ovarian, fallopian, pancreatic, prostate) and magnitude scaling with anomaly count.

Mechanism claim: The shared molecular substrate is HR-repair / FA-pathway wrapper failure. The shared molecular event is Knudson second-hit loss-of-heterozygosity, which can occur during embryonic morphogenesis (producing the developmental anomaly) and again in adult tissue (producing the cancer), in the same patient with the same first-hit germline variant.

Clinical claim: These predictions support precautionary cancer surveillance escalation today, before formal validation, because the cost asymmetry of being wrong in either direction favors action: missing a high-penetrance carrier costs late-stage presentation; over-screening a moderate-penetrance patient costs additional MRIs.

The rest of this paper derives all three claims from the channel hierarchy framework, marshals the existing evidence, identifies what is unstudied, defines the specific tests, and names the collaboration target.

2. The framework foundation: channel hierarchy and wrapper failure

The molecular foundation of this paper is the channel structure framework developed in McCarthy 2026 Papers 5 and 6. Briefly: cancer outcomes across patients are well-explained by “channel

When this second hit occurs in a cell of a hormone-responsive proliferative tissue (breast or ovarian epithelium), and the cell escapes apoptotic checkpoints, it becomes the founding cell of a clonal expansion that becomes a tumor. The cancer is the molecular consequence of two specific events in one cell: the inherited first hit and the somatic second hit.

The embryonic extension: the same molecular event (somatic second-hit LOH) can occur during embryogenesis. When it does, the affected cell is HR-deficient (not just half-functional but fully null) at the moment of the LOH event. If this happens in cells that are at that moment doing morphogenetic work — Müllerian duct fusion, metanephric blastema migration, neural tube closure, cardiac septation — the affected cell cluster fails to complete its morphogenetic program. The result is a localized developmental anomaly in exactly the tissue type and location where the LOH cells were doing structural work at the critical window.

Phenomenon	Knudson event	Tissue	Timing
Adult HBOC cancer in carrier	Stochastic second-hit LOH in adult breast/ovarian epithelium	Hormone-responsive proliferative tissue	Lifetime, cumulative; usually after age 30-40
Developmental anomaly in carrier	Stochastic second-hit LOH in embryonic morphogenetic cells	Tissue undergoing morphogenesis at the time of the LOH event	Embryonic critical window — usually weeks 4-12 of gestation, depending on the tissue

Müllerian duct fusion and metanephric blastema migration both occur in approximately weeks 6-10 of gestation. **A single cluster of LOH events affecting cells in that window could produce both Müllerian and renal anomalies in the same patient**, which is exactly the co-occurrence observed in approximately 7% of horseshoe kidney patients (per the urology literature). The framework reads this co-occurrence as a window-of-vulnerability signature: the two anomalies cluster because they share a developmental window and the LOH event population that fired in that window.

The unification: developmental anomaly and adult cancer in the same carrier are not two unrelated events that happen to coincide with a cancer-predisposing variant. **They are the same molecular event (Knudson second-hit LOH) at different timescales in different tissues of the same patient.** The patient who has experienced an embryonic LOH event (visible as an anomaly) has also demonstrated that their LOH machinery has “fired once” — and the framework predicts the second firing (in adult tissue, producing cancer) is more likely in such patients than in carriers who have not experienced the first firing.

Why this matters mechanistically: the prediction follows from a single molecular mechanism rather than from a vague “the gene matters in many tissues” hand-wave. It produces specific testable consequences:

1. **Anomaly-positive carriers’ tumors should show second-hit LOH at the same locus** — the same allele that was lost embryonically, lost again in the tumor cell. This is testable from tumor sequencing data of carriers with documented developmental anomalies.

2. **Multi-anomaly carriers should cluster their anomalies in the same developmental window** — because a single LOH event population fires within a narrow temporal range. Carriers with anomalies spanning multiple windows had multiple LOH events, which is rarer and predicts higher cumulative LOH rate.
3. **Anomaly-positive carriers should show elevated baseline LOH rates in non-tumor tissue** — clonal hematopoiesis (CHIP) at the relevant locus, telomere instability, or measurable LOH frequency in skin punch biopsies. Testable from existing CHIP cohorts if anomaly status is annotated.
4. **The polygenic background or environmental exposures that favor LOH should also predict both anomaly and cancer** — anomaly-positive carriers should have higher LOH-favoring polygenic risk scores.

None of these have been tested in published work. They follow from the framework’s mechanism and are the basis of the empirical case the paper defines in §6.

4. Existing evidence

Four lines of published evidence converge on the framework’s prediction without confirming it directly.

4.1 Bi-allelic FA-pathway loss → VACTERL-H + cancer (the homozygous extreme)

Patients with bi-allelic null or compound heterozygous null mutations in any FA-pathway gene develop Fanconi anemia, characterized by congenital anomalies (short stature, café-au-lait pigmentation, skeletal abnormalities, urogenital anomalies, cardiac defects, GI malformations, microcephaly) plus dramatically elevated cancer risk from infancy (AML, head and neck, gynecologic). Approximately 75% of FA patients show some congenital anomaly, with the constellation aligning closely with VACTERL-H. The PHENOS clinical criteria explicitly state: **“Thinking of VACTERL-H? Rule out Fanconi anemia”** [Alter et al].

The framework’s prediction is fully satisfied at the bi-allelic extreme. When both copies of the wrapper gene are lost from the start, the developmental anomalies and the cancer co-occur in the same patient by the same mechanism. The framework’s extension is that the same molecular logic should apply, in attenuated form, to heterozygous carriers when an embryonic second-hit LOH event occurs in critical morphogenetic cells.

4.2 Horseshoe kidney → 3-4× elevated cancer (single-anomaly precedent)

The urology literature documents that horseshoe kidney is associated with 3-4× elevated rates of certain renal cancers, including transitional cell carcinoma, Wilms tumor, and carcinoid tumors of the kidney. The standard explanation is anatomical (urinary stasis, abnormal positioning, ureteral compression). **The framework reads this as a marker of HR-pathway insufficiency during embryonic kidney migration, with the cancer risk reflecting the same molecular insufficiency rather than only the local anatomical conditions.** The two readings are not mutually exclusive, but the framework predicts an additional finding: horseshoe kidney patients should also show elevated cancer risk in non-renal tissues, especially HBOC-spectrum tissues. This non-local prediction has not been tested.

4.3 Daltveit et al 2020 BMJ — the inverted test, partially run, partially positive

Daltveit, Klungsøyr, Engeland et al 2020 BMJ, “Cancer risk in individuals with major birth defects: large Nordic population based case-control study among children, adolescents, and adults” [PMID 33268348] is the most directly relevant existing publication. Using Norwegian and Swedish national health registries, the study analyzed cancer risk in patients with major birth defects vs matched controls, across children (0-14), adolescents (15-19), and adults (20-46).

Result	Value
Overall birth defect → cancer OR	1.74 (99% CI 1.63-1.84)
Non-chromosomal structural defects	1.54 (1.44-1.64)
Chromosomal anomalies (mostly Down syndrome)	5.53 (4.67-6.54)
Children (0-14)	2.03
Adolescents (15-19)	1.58
Adults (20-46)	1.21 (still significant)
Defect categories included	eye, nervous system, urinary organs, skeletal dysplasia, genital organs, heart defects
Major specificity finding	“Many structural birth defects were associated with later cancer in the same organ system”

This is the inverted test, run at large scale, partially positive, and consistent with the framework’s broad prediction. Adults with major birth defects show OR 1.21 elevated cancer risk persisting from childhood at OR 2.03, with dose-response by anomaly count. The basic population-level signal that anomaly-positive patients have elevated cancer risk is published and replicated.

What Daltveit does NOT do, and what this paper proposes as the framework’s specific testable extensions:

1. **No HBOC-spectrum tissue specificity analysis.** Daltveit reports cancer in the “same organ system” — supporting the local-tissue / anatomical-stasis explanation. The framework predicts that, after adjusting for the local-tissue effect, anomaly-positive patients show elevated rates of HR-deficient cancer types (breast, ovarian, fallopian, pancreatic, prostate) in **distant** tissues. This finer-grained tissue-specificity test has not been run.
2. **No breast cancer receptor subtype analysis.** The framework’s mechanism predicts hormone-driven proliferation in HR-deficient cells, which is the BRCA2 / RAD51-paralog axis (more ER/PR+ than triple-negative). The framework specifically predicts that the breast cancer elevation in Müllerian anomaly patients should concentrate in ER/PR+ rather than triple-negative subtype.
3. **Age cap at 46 underestimates HBOC cancers.** Breast cancer (especially the ER/PR+ subtype) and ovarian cancer often present after 50. Daltveit’s adult cohort (20-46) is too young to capture the full lifetime HBOC signal. **The OR 1.21 in adults is plausibly an underestimate of the lifetime effect for HBOC-spectrum cancers specifically.**
4. **No mechanism.** Daltveit reports the descriptive epidemiology and notes the same-organ-system pattern. The framework provides the molecular mechanism (embryonic Knudson

second-hit LOH) that turns the descriptive epidemiology into a testable causal claim with specific molecular predictions.

4.4 Open Targets Platform — gene-disease aggregation

We queried the Open Targets Platform GraphQL API for HBOC gene $\rightarrow$ disease associations (BRCA1, BRCA2, PALB2, BRIP1, RAD51C, RAD51D). Reproducible script and raw responses in `analysis/paralog_projection/query_opentargets_hbo_developmental.py` and `data/opentargets/`. Two findings of note for this paper:

- **Every HBOC gene has its corresponding Fanconi anemia complementation group as a top association**, with genetic-association scores 0.46-0.84. The bi-allelic developmental syndrome is firmly published at the gene-disease level for every gene tested, driven by clinical-genetics submissions (eva / ClinVar) rather than literature mining.
- **BRCA2 $\rightarrow$ esophageal atresia / tracheoesophageal fistula at genetic-association score 0.809; BRIP1 $\rightarrow$ same at 0.559; BRCA2 $\rightarrow$ Joubert syndrome (cerebellar malformation) at 0.915.** These are specific VACTERL-spectrum or developmental phenotypes scored as top associations for these genes. The signals come from eva (ClinVar pathogenic variant submissions), not literature mining or animal models. **The molecular link “HBOC gene $\rightarrow$ specific VACTERL-spectrum developmental phenotype” is firmly published at the gene-disease aggregation level.**

These aggregations cannot distinguish bi-allelic from heterozygous causation (Open Targets does not separate by zygosity), so the EA/TEF and Joubert signals are likely driven mostly by bi-allelic FA-D1 / FA-J case reports. But the molecular link is no longer speculation. The bi-allelic literature scaffolds the mechanism the framework’s heterozygous extension predicts.

4.5 Adjacent territory — what is in the literature but does not test the framework

For completeness:

- **Uterine fibroids $\rightarrow$ breast cancer HR 1.31** [Kang 2017]. Fibroids are acquired (leiomyoma), not congenital. Different mechanism. Not a test of congenital Müllerian $\rightarrow$ breast cancer.
- **Endometrial-breast bidirectional Mendelian randomization** showing causal relationship between two adult cancers. Different question: two adult cancers, not developmental anomaly $\rightarrow$ adult cancer.
- **Uterine carcinosarcoma over-represented in breast cancer patients** — co-occurrence of two cancers, with “possibly genetic component” noted but unexplained.
- **Müllerian anomaly classifications (ASRM 2021)** — anatomy and fertility, no cancer follow-up.
- **Standard breast cancer risk calculators (Gail, BCRAT, IBIS)** — do NOT include congenital uterine anomaly status as a risk variable.

The specific prediction “Müllerian congenital anomaly $\rightarrow$ ER/PR+ breast cancer at the population level” has not been published, tested, or even seriously proposed in the existing literature. This is the cleanest novel testable claim the framework makes, and it is the entry point of paper 8c.

5. The literature gap, sharpened

What is not yet published, in order from most specific to most general:

1. **Müllerian developmental anomaly $\rightarrow$ ER/PR+ breast cancer in distant tissue** — the framework’s load-bearing specific prediction, with no precedent in the literature.
2. **HBOC-spectrum tissue specificity test** — does the Daltveit OR 1.21 in adults concentrate in HR-deficient cancer types when broken out by ICD-10 site code? Not in any published analysis.
3. **Cross-tissue (non-local) elevation analysis** — Daltveit emphasizes same-organ-system. The framework predicts the elevation extends beyond local tissue. The cross-tissue analysis has not been published.
4. **Dose-response by anomaly count restricted to HBOC-spectrum cancers** — Daltveit reports dose-response for any cancer; the HBOC-specific dose-response has not been broken out.
5. **Knudson second-hit LOH testing in tumors of anomaly-positive carriers** — testable from existing tumor sequencing data with appropriate phenotype annotation; not published.
6. **CHIP / clonal hematopoiesis frequency in anomaly-positive carriers** — testable from existing CHIP cohorts; not published.
7. **Variant-stratified penetrance modification** — the carrier-stratified version that requires linked genotype data; not published, more difficult to access.

Items 1-4 are the framework’s most actionable testable extensions. They are testable on existing Norwegian/Swedish registry data with the addition of cancer-subtype information that is already present in those national registries. The Daltveit team’s existing infrastructure could run these analyses by re-querying the same dataset they already published on.

6. The specific test: Müllerian $\rightarrow$ ER/PR+ breast cancer

The cleanest test the framework supports is the specific Müllerian $\rightarrow$ ER/PR+ breast cancer prediction. We define it operationally:

Inclusion criteria: - Female patients with at least one ICD-10 Q-code in Q51 (congenital malformations of uterus and cervix, including Q51.3 bicornuate, Q51.2 other doubling, Q51.4 unicornate, Q51.6 embryonic remnants, Q51.7 communication between uterus and digestive/urinary tracts, Q51.8 other) or Q52 (other congenital malformations of female genitalia). - No requirement for genetic testing.

Comparison: age-, race/ethnicity-, parity-, age-at-first-birth-, BMI-, hormone-exposure-matched controls drawn from the same registry without any Q51/Q52 code.

Primary outcome: incidence of breast cancer (ICD-10 C50) before age 75, stratified by hormone receptor subtype: - ER+/PR+/HER2- (“luminal A”) - ER+/PR+/HER2+ (“luminal B”) - ER+/PR-/HER2 $\pm$ (other ER+) - ER-/PR-/HER2- (triple-negative) - ER-/PR-/HER2+ (HER2-enriched)

Secondary outcomes: ovarian (C56), fallopian (C57), uterine (C54-C55) cancers in the Müllerian-anomaly cohort vs controls. (Uterine cancer is the local-tissue effect Daltveit’s same-organ-system finding would predict; the framework adds the distant-tissue prediction for breast.)

Tertiary outcome: dose-response by Müllerian anomaly severity (uni/bi/septate vs full uterus didelphys vs combined uterine + vaginal + cervical anomalies).

Predicted result (the framework’s specific claim): - **SIR / HR > 1.5 for ER/PR+ breast cancer subtypes** in the Müllerian-anomaly cohort vs matched controls. - **Effect predominantly in ER/PR+ subtypes** (luminal A and luminal B), with attenuated or null effect in triple-negative. - **Dose-response with anomaly severity** — multi-feature anomalies show larger elevation than isolated single-feature anomalies. - **Effect persists after adjusting for parity, age at first birth, and hormone exposure** (the standard breast cancer confounders). - **Effect persists after excluding patients with known HBOC family history or genetic test positivity** — the framework’s prediction is independent of known carrier status because it reflects underlying wrapper insufficiency that may not have been molecularly characterized.

Falsification conditions: - $\text{SIR} \approx 1.0$ (95% CI containing 1.0) $\rightarrow$ framework’s specific prediction fails. - Effect concentrated in triple-negative subtype rather than ER/PR+ $\rightarrow$ mechanism interpretation is wrong; the connection is via a different molecular pathway than HR-deficient hormone-responsive proliferation. - Effect entirely explained by reproductive confounders $\rightarrow$ the anomaly is a marker of reproductive history, not of substrate failure. - No dose-response $\rightarrow$ the framework’s “more wrapper failure $\rightarrow$ more LOH events” prediction fails. - Effect present only in patients with confirmed HBOC variants $\rightarrow$ the framework’s “mechanism applies regardless of variant testing” prediction fails (this would still partially confirm a more limited carrier-specific version, but would falsify the population-level extension).

Power: SEER and major US state cancer registries record both ICD-10 Q-codes and breast cancer receptor subtype. UK Biobank and the Nordic registries provide the same. Even narrow Müllerian Q-code definitions yield thousands of cases in million-person datasets. Statistical power for detecting $\text{HR} \approx 1.5$ is ample.

The same logic generalizes to other VACTERL-spectrum components: cardiac anomaly Q20-Q24 $\rightarrow$?, renal anomaly Q60-Q64 $\rightarrow$?, esophageal atresia Q39 $\rightarrow$?. The framework’s broader prediction is that each developmental anomaly type predicts elevation in HR-deficient cancers in tissues that share embryonic developmental windows or hormonal/proliferative similarity. The Müllerian $\rightarrow$ ER/PR+ breast cancer prediction is the cleanest entry point because the receptor subtype concentration is a sharp, falsifiable, mechanism-specific test.

7. The Nordic HBOC cohort as the natural collaboration target

The Daltveit team is the natural collaboration target for paper 8c’s specific test. They have already run the basic inverted analysis at large scale on Norwegian and Swedish national health registries, producing the OR 1.74 birth defect $\rightarrow$ cancer signal. They have institutional access, IRB approvals, established methodology, and the same registry infrastructure that contains both the Q-codes and the cancer outcomes the framework’s specific test requires.

What the Nordic registries plausibly contain that has not been analyzed:

- Q51/Q52 Müllerian anomaly cases linked to long-term breast cancer follow-up
- Cancer subtype (receptor status) data, present in both Norwegian and Swedish national cancer registries
- Adequate sample size to break out the HBOC-spectrum tissues and the receptor subtypes

- Population-level genetic data via BRCA test results recorded in clinical genetics consultations (Norwegian BRCA testing has been routine for several decades, and a national overview of BRCA1 variants exists [Heramb et al 2021])
- Norwegian RAD51C carrier identification — at least one published Norwegian breast and ovarian cancer cohort identified pathogenic variants in seven HBOC genes including RAD51C
- The ability to run the analysis stratified by genotype OR un-stratified by genotype, depending on which subset is available

Why the Nordic cohort specifically:

1. The Daltveit team has already done the heavy lifting to build the linkage infrastructure between birth records, congenital anomaly registries, and cancer outcomes.
2. The Norwegian and Swedish national health systems have decades of high-quality structured records, with national ID linkage permitting longitudinal follow-up across all registries.
3. The HBOC clinical genetics community in Norway is well-organized, with national variant interpretation consortia (the BRCA1 Norway initiative being one example) and established clinical genetics services.
4. The Norwegian/Swedish populations contain documented carriers of HR-repair pathway pathogenic variants, including the variant the framework’s motivating single-N case (§9) carries. The variant came through Norwegian/Swedish maternal lineage in that case, which is consistent with the variant being present in measurable numbers in the Nordic populations the test would run on.

The collaboration ask: re-analyze the existing Daltveit dataset for HBOC-spectrum tissue specificity, ER/PR receptor subtype concentration in breast cancer outcomes, and anomaly-type-specific cross-tissue elevation. This is a re-analysis, not a new data collection, and could plausibly produce a follow-up paper to Daltveit 2020 BMJ within months rather than years.

8. The clinical action that follows today

The combined existing evidence is already enough to support precautionary clinical action, even before the framework-specific tests in §6 run. The supporting evidence in aggregate:

- **Daltveit 2020 BMJ:** OR 1.74 for any major birth defect → adult cancer in millions of Nordic records, with adult-period OR 1.21 still significant.
- **Bi-allelic Fanconi anemia → VACTERL-H + childhood cancer:** fully documented at the molecular level for every FA-pathway gene.
- **Horseshoe kidney → 3-4× elevated kidney cancer:** published urology literature, twenty years old.
- **Open Targets gene-disease:** HBOC genes are linked to specific VACTERL-spectrum developmental phenotypes via ClinVar pathogenic variant submissions at high genetic-association scores.
- **Cost asymmetry:** missing a high-penetrance anomaly-positive patient produces late-stage cancer presentation. Over-screening a moderate-risk anomaly-positive patient produces additional MRIs.

The recommendations the framework supports today:

For oncologists, primary care physicians, gynecologists, and genetic counselors:

Any patient with a documented history of a Müllerian (Q51, Q52) or VACTERL-spectrum (Q05, Q39, Q42, Q20-Q24, Q60-Q64, Q66-Q74, Q03) congenital anomaly, regardless of whether they have undergone genetic testing, should be treated as having moderately elevated baseline risk for HBOC-spectrum cancers (breast, ovarian, fallopian, prostate, pancreatic). For Müllerian-anomaly patients specifically, this means earlier first breast MRI screening (consider starting at 35 rather than 40) and more frequent monitoring than the standard population schedule. For multi-anomaly patients, the recommendation is stronger.

For HBOC genetic testing decisions:

Patients with two or more VACTERL-spectrum or Müllerian anomalies, even without family history of cancer, should be offered HR-deficient gene panel testing as part of routine workup. The threshold for offering panel testing should be lower for these patients, because the framework predicts that the anomaly is itself an indicator of underlying wrapper insufficiency that may have a genetic basis worth identifying for both the patient and their family.

For known HBOC variant carriers:

Carriers who additionally have documented Müllerian or VACTERL-spectrum congenital anomalies should be considered for the high-intensity end of the surveillance interval recommended for their gene, including earlier first MRI, more frequent monitoring, and earlier consultation for risk-reducing surgery. The patient should be told that the framework supports treating the anomaly as informative about realized penetrance, that the recommendation reflects mechanism-based reasoning grounded in the published epidemiology, and that the recommendation will be revised as the framework-specific tests in §6 produce actual subtype-stratified effect-size estimates.

Boundaries: - This is precautionary surveillance, not diagnostic certainty. The recommendation increases monitoring intensity, not the threshold for cancer diagnosis. - The recommendation does NOT support invasive intervention (prophylactic surgery, chemoprevention) on the basis of congenital anomaly status alone. - The recommendation can be revised — relaxed if the §6 tests produce null results, strengthened if they produce $SIR > 2.0$ or identify particularly high-risk subsets. - The single exception that already exists in current standard of care is Down syndrome, where elevated leukemia surveillance is routine. The framework’s claim is that this exception should be generalized to other anomaly types based on the existing epidemiology.

9. The motivating single-N case (Tricia)

This paper was motivated by a specific patient. With permission, the patient is Tricia, the wife of one of the authors. She is a heterozygous carrier of the RAD51C splice variant **c.1026+5_1026+7delGTA** — a deletion at the splice donor of intron 8 producing aberrant splicing including exon 8 skipping (Sanoguera-Miralles et al 2020, classified as pathogenic with ~98% Bayesian probability). The variant came through her maternal lineage; her mother is Norwegian/Swedish.

Her medical history includes a **bicornuate uterus** (Müllerian developmental anomaly — incomplete Müllerian duct fusion during embryonic weeks 6-10) and a **horseshoe kidney** (renal develop-

mental anomaly — failed migration of metanephric blastemas during the same embryonic window). She would not have survived to adulthood without modern pediatric urologic intervention. As an adult, she developed an aggressive HBOC-spectrum cancer that progressed in standard treatment but showed dramatic clinical response to the next-generation PARP1-selective inhibitor AZD5305 (saruparib) in the PETRA trial. At the time of her enrollment, she was one of fewer than 10 RAD51C carriers worldwide treated with the drug; in the PETRA trial dose-escalation phase she was one of one (61 patients total, only 1 with a RAD51C mutation).

She is N=1, NOT the basis of the framework’s prediction. The prediction is derived from the channel hierarchy framework in Papers 5 and 6, not extrapolated from her case. The Daltveit BMJ paper provides the population-level evidence at large scale. The Knudson embryonic-second-hit mechanism (§3) provides the molecular bridge. **Her case is consistent with the framework’s prediction but is not evidence for it.** In the framework’s own methodological commitment to proof by elimination across many independent projections, no single patient is load-bearing.

What her case illustrates, when viewed through the framework, is the mechanism the paper predicts:

- The bicornuate uterus and horseshoe kidney are not two unrelated developmental coincidences. Both are failures of fusion/migration events in the same embryonic window (weeks 6-10 of gestation). The framework reads them as embryonic Knudson second-hit LOH events in cells doing urogenital morphogenesis at that window.
- The adult HR-deficient breast/ovarian cancer is another Knudson second-hit, in a different tissue, decades later, on the same first-hit germline variant.
- Both events are mechanistically the same. Modern pediatric urologic surgery fixed the structural consequences of the first event but did not fix the underlying wrapper insufficiency. The second event was therefore predictable in principle, even if not in detail.
- The dramatic response to AZD5305 reflects that her tumor’s HR pathway is broken at a routing/scaffold layer (RAD51C) where the parallel BER pathway blockade is the surgical fix — the wrapper-routing-failure response pattern documented in the Tricia anchor case memory and the wrapper variance hypothesis.

If the framework’s prediction is right, she is one carrier of a variant that exists in measurable numbers in the same Nordic populations the §6 tests would run on. The framework’s claim is that her case is not unusual but unstudied — that the population-level analysis of Müllerian-anomaly carriers will identify a measurable subset of similar carriers whose clinical management should differ from the standard. Her single-N case does not validate that claim. The §6 test would.

The motivation is personal. The prediction is not.

10. What this paper is not

We are explicit:

- **It does not claim that the basic inverted test is novel.** Daltveit 2020 BMJ already runs that test at large scale. The framework’s contributions are the mechanism, the HBOC-spectrum tissue specificity extension, and the receptor subtype prediction.
- **It does not claim that congenital anomalies cause cancer.** The anomaly is predicted to be a marker of wrapper-failure pleiotropy, not a cause. The cancer is caused by the same

molecular mechanism that produced the anomaly, not by the anomaly itself.

- **It does not claim that all anomaly-positive carriers will develop cancer.** Penetrance is not 100%. The claim is about elevated conditional probability relative to anomaly-negative carriers.
 - **It does not claim that current standard of care is wrong.** Current management is reasonable given current evidence. The paper claims that current evidence is incomplete in a specific way, identifies the test that would close the gap, and supports a precautionary recommendation in the meantime.
 - **It does not claim that the §6 test will succeed.** The test is falsifiable per §6.
 - **It does not claim that the §8 surveillance recommendation is established standard of care.** It is precautionary, mechanism-based, and revisable.
 - **It does not claim that the framework is universally true for all cancer predisposition syndromes.** The claim is specifically about HR-repair / FA-pathway / wrapper-class genes. Whether the same logic applies to Lynch (MMR), Li-Fraumeni (TP53), Cowden (PTEN), or other syndromes is a separate question that the same methodology could address.
 - **It does not claim that Tricia is evidence.** She is the motivating N=1 illustration. Her case is consistent with the framework’s prediction, not load-bearing for it.
 - **It does not claim that “positive thinking cures cancer,” “you gave yourself cancer by being stressed,” or any other version of psychogenic cancer attribution.** The framework’s claim is about molecular wrapper failure observable in cellular and developmental biology, not about psychological or behavioral causation.
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11. Why this matters

The number of patients with documented Müllerian anomalies or VACTERL-spectrum anomalies is not large in absolute terms. Müllerian malformations occur in approximately 5-7% of women in the general population (depending on definition); horseshoe kidney occurs in approximately 1 in 500. The combined VACTERL-H-M cohort the framework predicts as elevated-risk is a few percent of the population. This is a small fraction in proportional terms.

But the cost of treating these patients as standard-risk is high. **The patients most likely to benefit from earlier and more frequent surveillance are exactly the patients the current model treats as having no special risk.** Adult women with bicornuate uterus discovered during fertility workup are not currently flagged for early breast MRI. Patients with horseshoe kidney discovered incidentally on imaging are not currently flagged for HBOC panel testing. Adults with repaired esophageal atresia from infancy are not currently flagged for any cancer surveillance escalation. The framework’s prediction is that all of these omissions are leaving treatment on the table for a measurable fraction of patients.

The cost asymmetry favors action. Additional MRI for an anomaly-positive woman who turns out to be moderate-risk costs the cost of an MRI. Missing a high-risk anomaly-positive woman whose cancer presents at stage 3 instead of stage 1 costs years of life and dramatic treatment escalation. **Even at the OR 1.21 effect size from the Daltveit adult cohort, the precautionary recommendation is justified by the cost structure alone, before the framework-specific tests refine the effect size further.**

The translational gap between the published epidemiology and current clinical practice is the gap this paper proposes to close. The Daltveit-style finding has been in print for years. The bi-allelic

FA evidence has been in print for decades. The horseshoe-kidney $\rightarrow$ cancer literature has been in print for twenty years. **None of these has produced a clinical recommendation to treat anomaly-positive patients as elevated baseline cancer risk, outside of Down syndrome.** This paper argues that the gap should close, and that the framework provides both the mechanism and the specific testable extension that justifies closing it.

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Status: Positional paper, draft v3. Checkpoint. The position is staked, the existing evidence is acknowledged, the framework-specific extensions are defined, the collaboration target is named, and the clinical action is laid out.

The core revisions from v2: - Acknowledged Daltveit 2020 BMJ as the existing inverted test, dropping the claim that the basic inverted test is novel - Added §3 introducing the Knudson embryonic-second-hit mechanism as the molecular bridge - Reframed the position in §1 around the specific Müllerian → ER/PR+ breast cancer prediction - Rewrote §6 to focus on the receptor-subtype-stratified test rather than the general inverted test - Renumbered: collaboration target (Nordic cohort) is now §7, clinical action is §8, the single-N case is §9, what-this-is-not is §10, why-this-matters is §11 - §9 (Tricia) explicitly framed as N=1 motivating illustration, NOT as evidence; the prediction is derived from the channel hierarchy framework - Added the variant identity, splice mechanism, ancestry, and PETRA trial cohort context to §9 - Strengthened §8 clinical recommendations with the OR 1.74 backing from Daltveit - Maintained the convergence-first / mystery-novel-inverted methodology throughout

The paper is now ~6500 words. It is intentionally short for a positional paper because it makes one specific claim, marshals the evidence, and gets out of the way of the test that resolves it.

Reproducibility: the OpenTargets query script and raw GraphQL responses (`analysis/paralog_projection/queries` and `data/opentargets/`) reproduce the §4.4 evidence. The reproducibility notebook for the broader paper is at `notebooks/paper8b_reproduce.py` and serves as the template for an analogous `paper8c_reproduce.py` (TODO).